



CNC Machine Tool Programming

Groover M.P., Automation, Production Systems, and Computer-
Integrated Manufacturing: Chapter 7



The student knows:

- How to program CNC machine tools;
- What the benefits of CAD/CAM integration are.



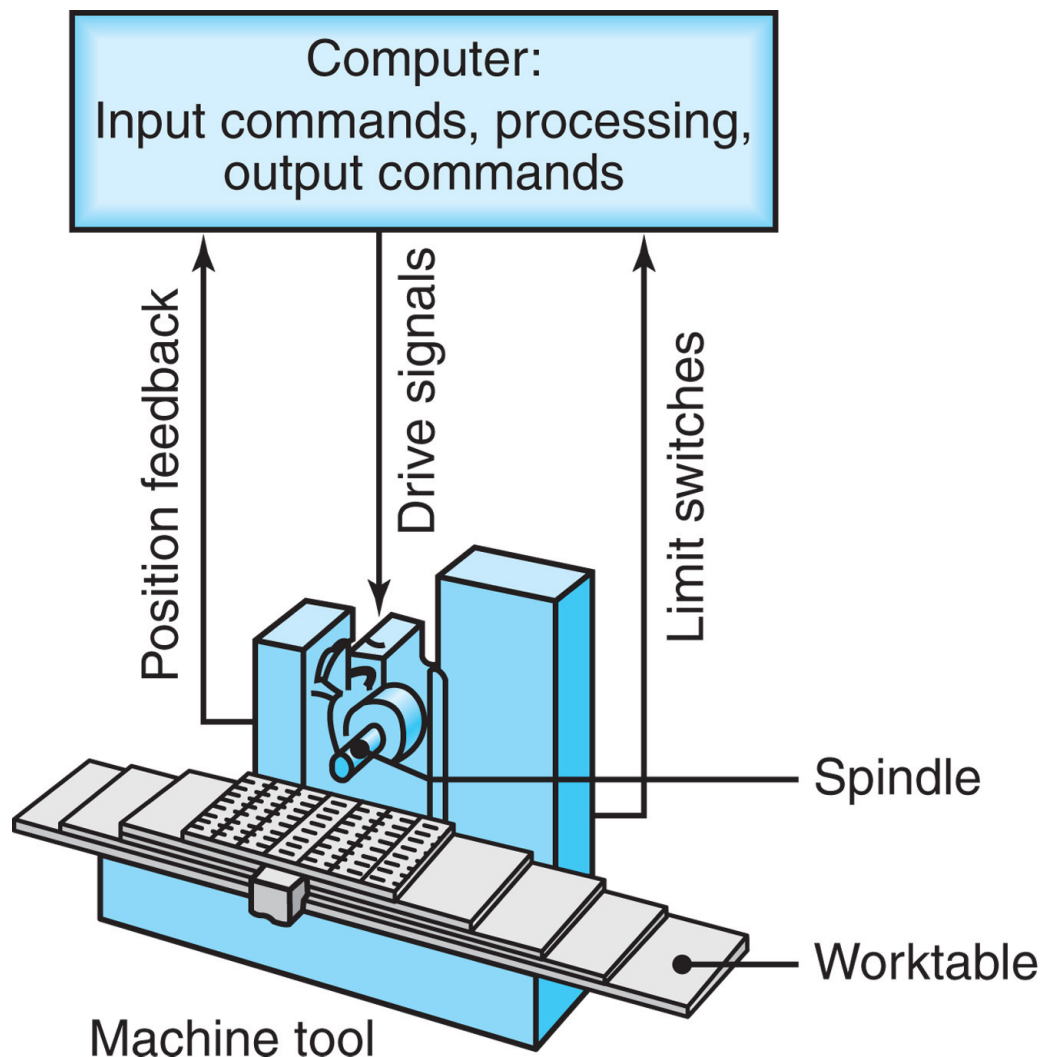


Numerical Control

Automatic control of a process performed by a device that makes use of numerical data introduced while the operation is in progress (ISO 2806:1994)

Computer Numerical Control

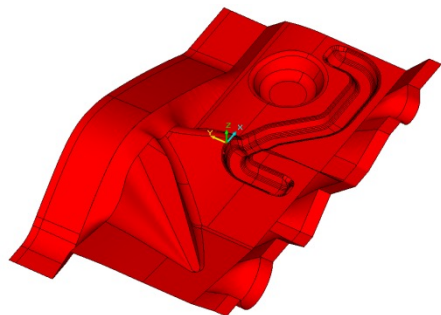
Realization of NC using a computer to control the machine functions (ISO 2806:1994)



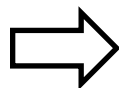


MACHINE TOOL PROGRAMMING

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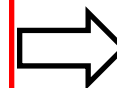


Component

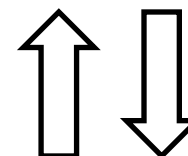


```
***  
G01    X0.0    Y0.0;  
G01    X2.0    Y0.0;  
G01    X2.0    Y2.0;  
G01    X0.0    Y2.0;  
G01    X0.0    Y0.0;  
***
```

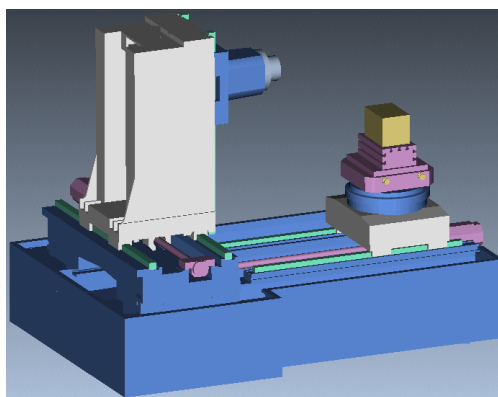
Part program



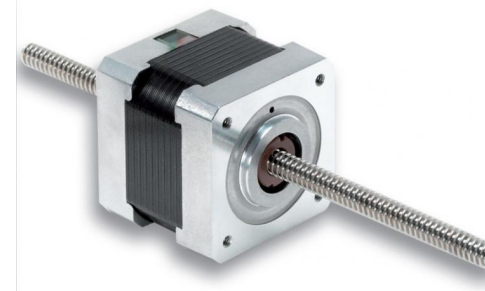
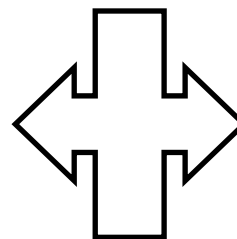
Numerical control



Signals
Feedback



Machine tools
Fixtures
Tools



Positioning System

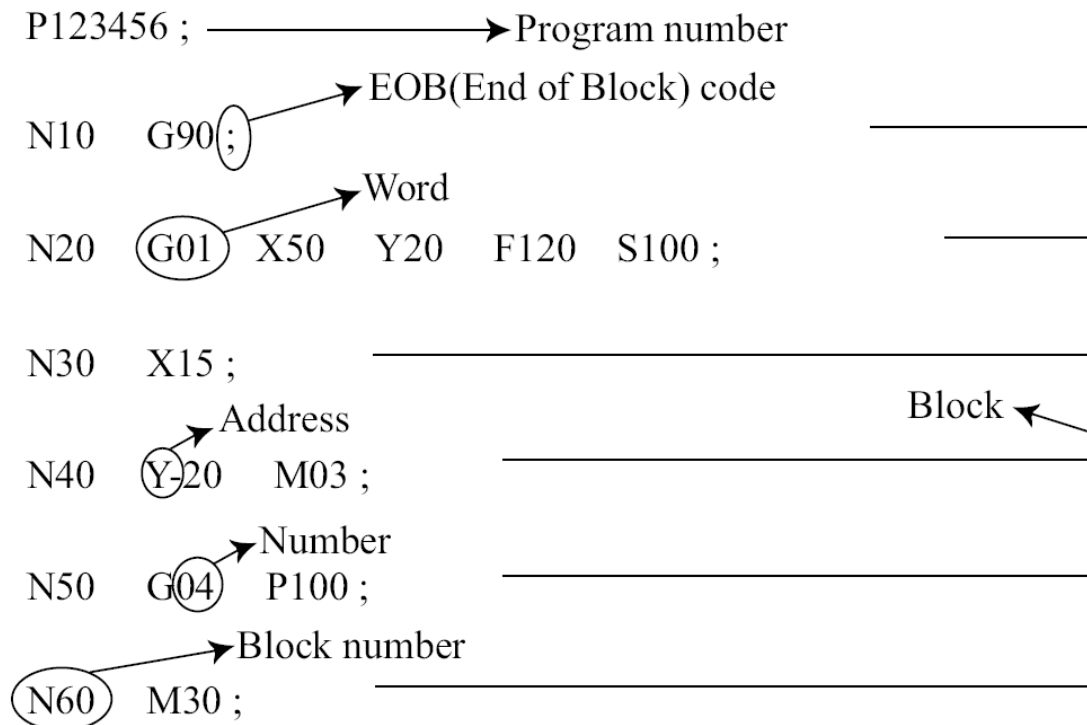


*Ordered set of instructions in a language and in a format required to cause operations to be effected under automatic control, which is either written in the form of a **machine program** on an input medium or prepared as input data for processing in a computer to obtain a machine program (ISO 2806:1994).*



PART PROGRAM: ISO CODE

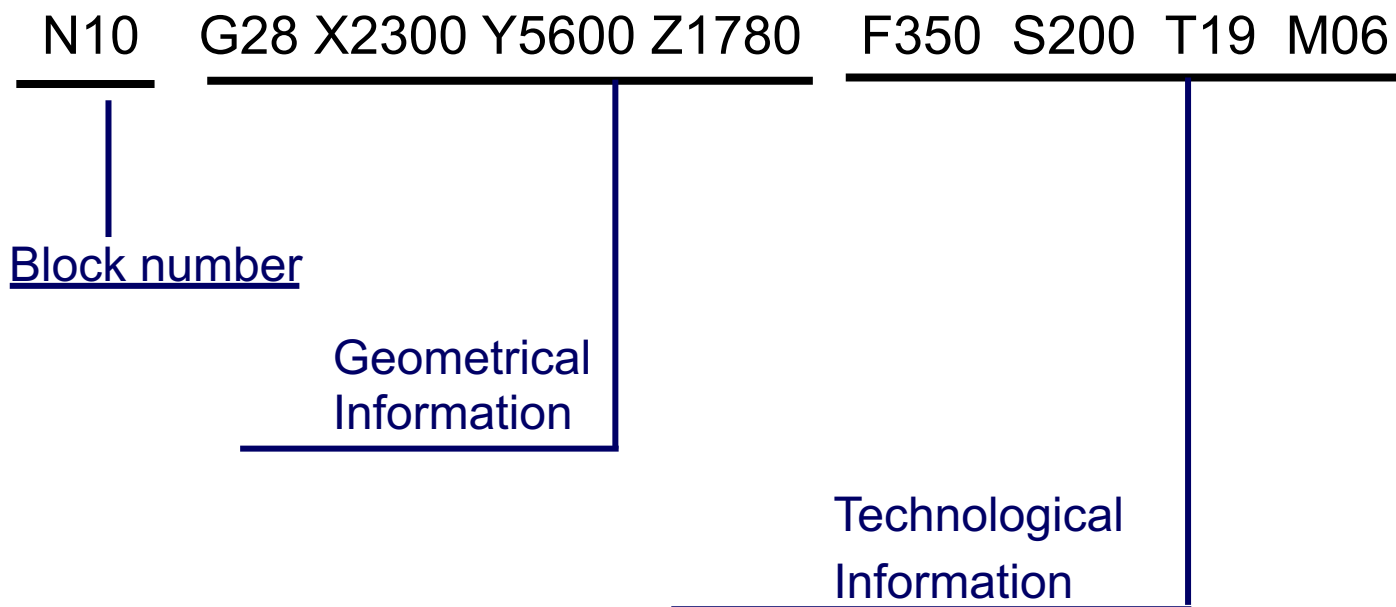
- NC commands such as G, M, S, T, H, D, F code and related address
- Call of sub-program and displaying message
- Setting variable and conditional program calls





BLOCK EXAMPLE

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Motion

A, B, C, U, V, W, X, Y and **Z**

Addresses

Machine functions

F, G, M, N, S and **T**

A	B	C	D	E	F	G	H	I	L	M	N
O	P	Q	R	S	T	U	V	W	X	Y	Z

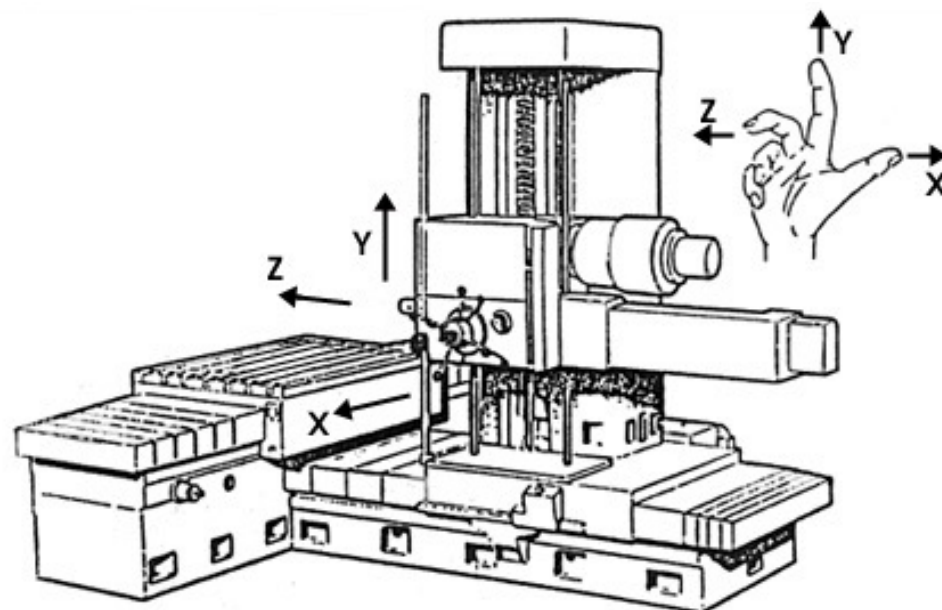
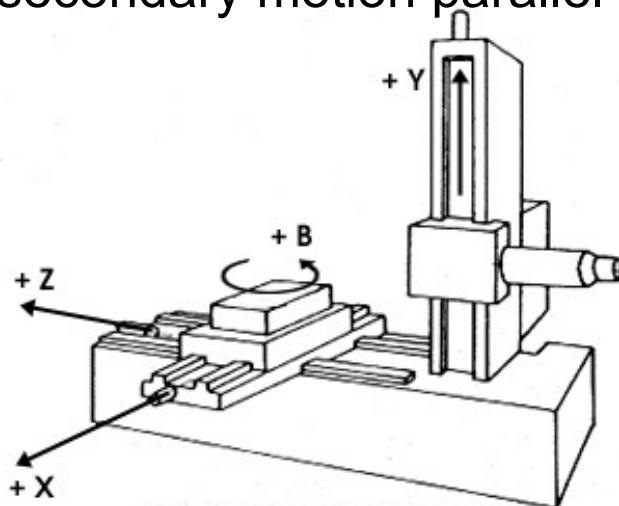
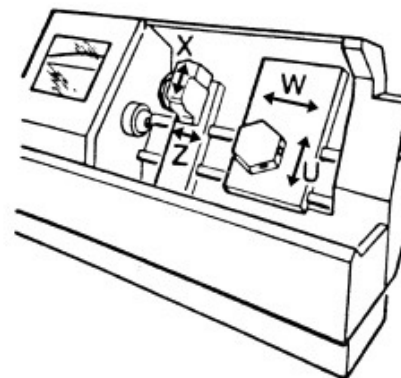
Modal: effective throughout the following blocks until the modal cancel command is used.

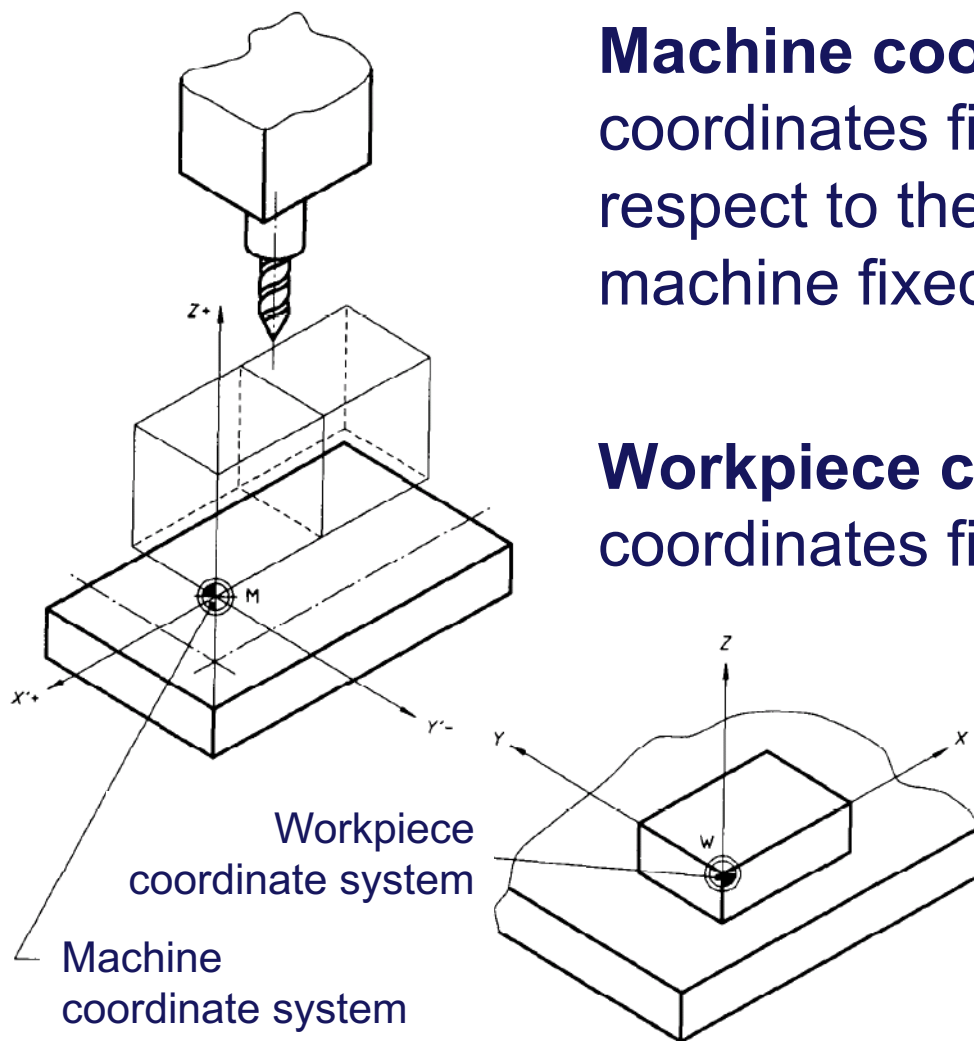
Non-modal: effective within the commanded block and automatically cancelled by the next block.



MOTION ADDRESSES

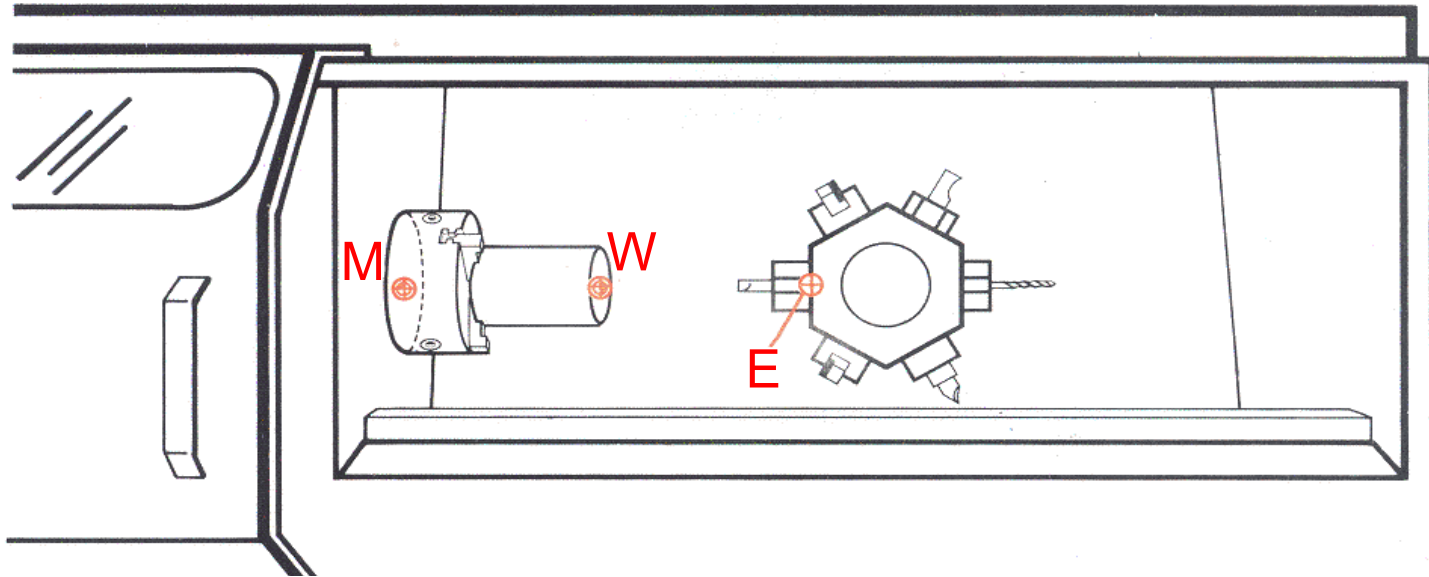
- X** main motion of X linear axis
- Y** main motion of Y linear axis
- Z** main motion of Z linear axis
- A** angular coordinate around X
- B** angular coordinate around Y
- C** angular coordinate around Z
- U** secondary motion parallel to X
- V** secondary motion parallel to Y
- W** secondary motion parallel to Z





Machine coordinate system: Cartesian coordinates fixed on a machine with respect to the machine zero (origin of the machine fixed by the manufacturer).

Workpiece coordinate system: Cartesian coordinates fixed on the workpiece.

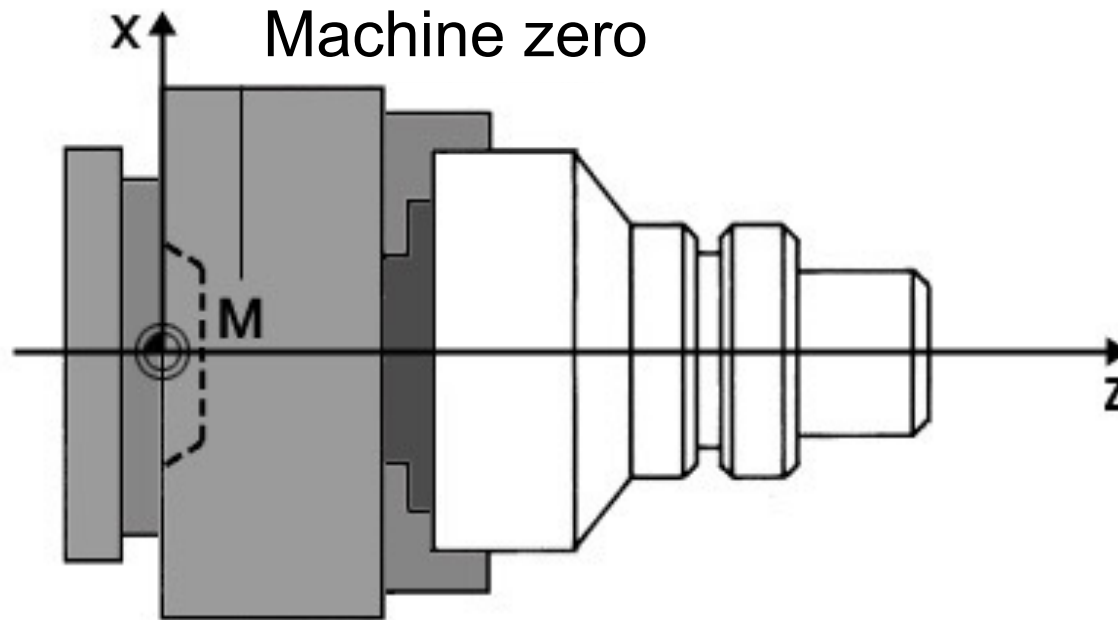


- **M**: machine coordinate origin (machine zero)
- **W**: workpiece coordinate origin (part zero)
- **E**: tool coordinate origin (tool zero)



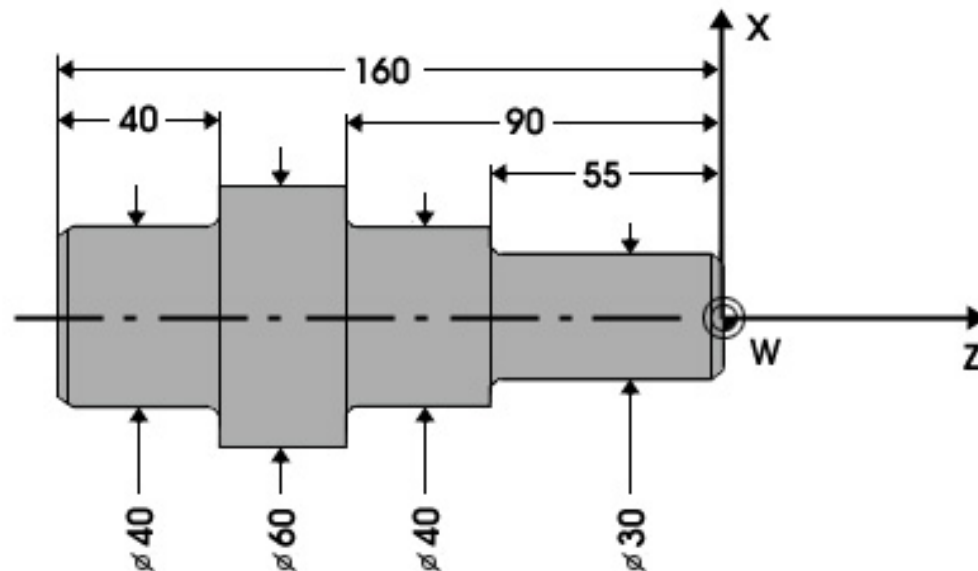
MACHINE ZERO IN TURNING CENTERS

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Selected on the basis of:

- Technical drawing
- Convenience of setup and operation
- Safety of working conditions





PREPARATORY FUNCTIONS (G-CODE)

- **Movement:** commands to define the relative movement between tool and workpiece.
- **Setting local coordinate system:** commands to specify the origin and orientation of local coordinate system including orthogonal coordinate system, the polar coordinate system, and rotational coordinate system.
- **Interpolation:** commands to cut various profiles such as line, arc, helical, spline, etc.
- **Miscellaneous:** commands to define tool compensation, safety check, etc.



AUXILIARY FUNCTIONS (M-CODE)

- **Coolant:** coolant on or off.
- **Start/End** of a main and sub- program: beginning and ending of a part program.
- **Spindle:** commands to let the spindle to rotate in CCW or CW direction and specify the spindle speed, limit speed and spindle gear change.
- **Miscellaneous:** commands to change the tool, the pallet, etc.



It can be the only instruction in the block, three (3) is the limit on the number of M instructions in a block. It is activated before the movement commanded in the same block (G instruction).

N10 G00 X-60 Y-20 S500 M03 M08
 └──────────┘
 movement



M00	stop program
M03	CW spindle rotation
M04	CCW spindle rotation
M05	Stop spindle rotation
M06	Automatic Tool Change
M08	Coolant on
M09	Coolant off
M72	Tailstock block
M10/11	Chuck open/close (lathe)
M60	Automatic Pallet Change
M30	End of the program

...



SPINDLE CONTROL (S COMMAND)

- The address S relates to machine spindle function
- Two alternatives:
 - Cutting speed in [m/min] – G96
 - Spindle speed in [rpm] - G97
- The spindle rotation direction must be specified:
 - Clockwise (CW) – M03
 - Counterclockwise (CCW) – M04
- Spindle will not start rotating until both the speed and the rotation commands have been processed
- Stop the spindle rotation – M05
- Other M-commands stop the rotation (M00, M01, M02, M30, etc.)



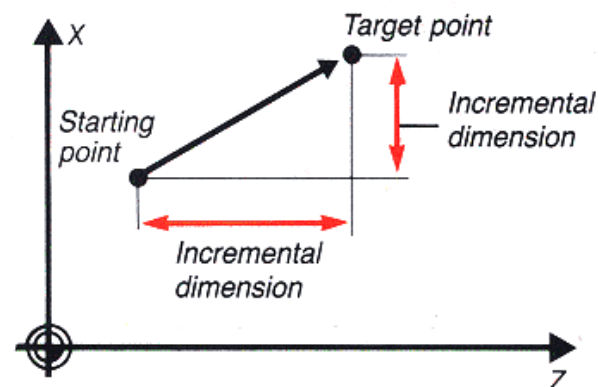
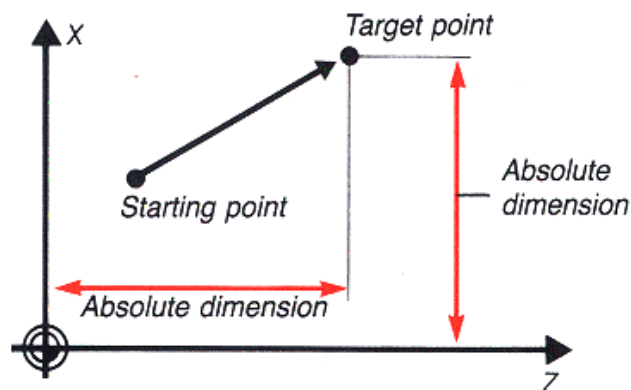
FEED RATE CONTROL (F COMMAND)

- The address F relates to how fast the tool will move
- Two alternatives:
 - Feed rate per minute [mm/min] – G94
 - Feed rate per revolution [mm/rev] - G95
- Maximum programmable cutting feed rate is determined by machine manufacturer
- While running a program, the programmed feed rate may be temporarily suspended or changed by *feedhold override switch* and *feed rate override switch*



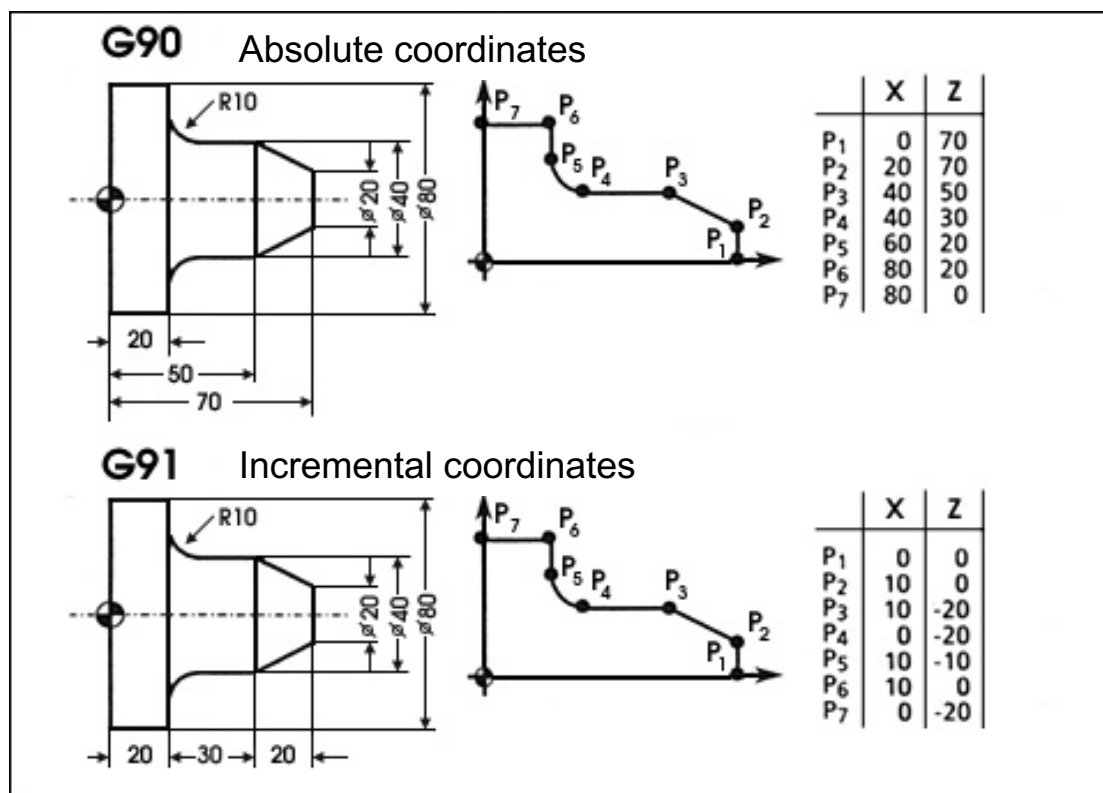
G90: absolute coordinate system

G91: incremental coordinate system





In turning absolute quotes are on the diameter, incremental on the radius.

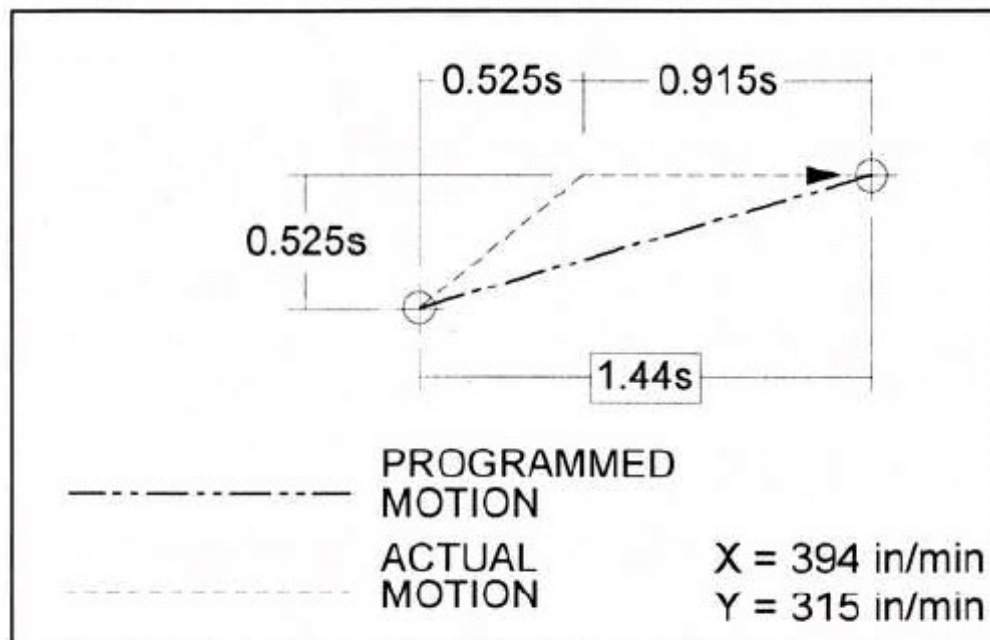




RAPID POSITIONING (G00)

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The rapid movement function (G00) is used for commanding the specific axes to move rapidly to the programmed position.



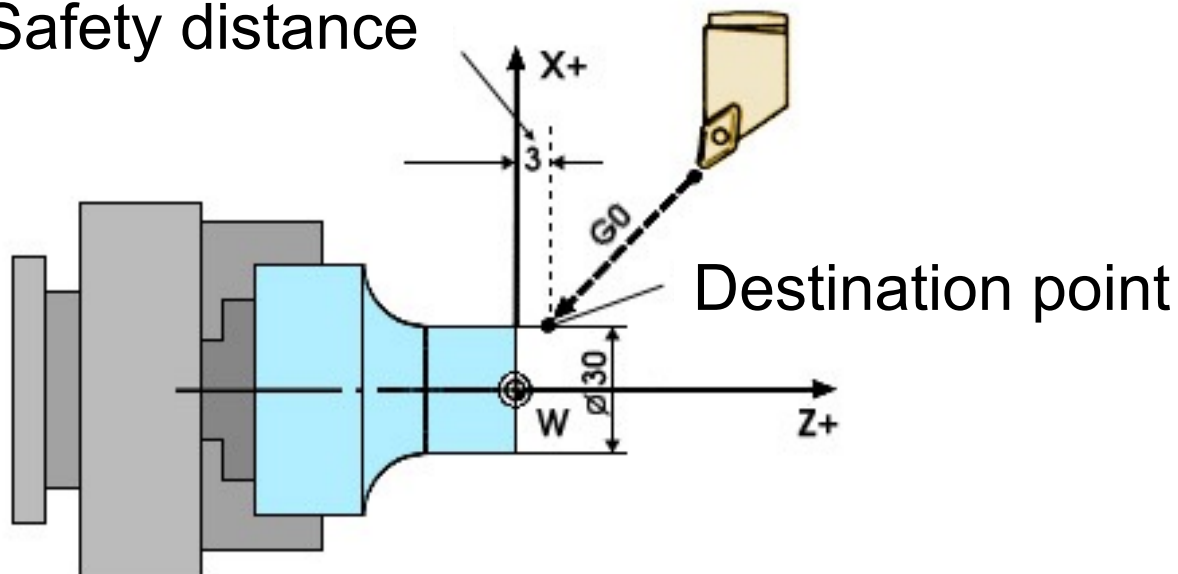
Different rapid rate for each axis



EXAMPLE OF POSITIONING

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Safety distance

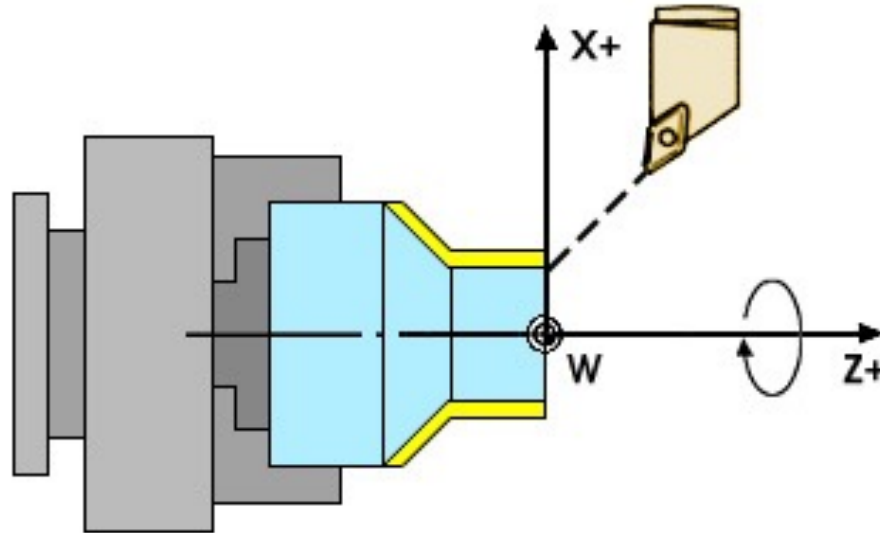


N100 G00 X30 Z3



LINEAR INTERPOLATION (G01)

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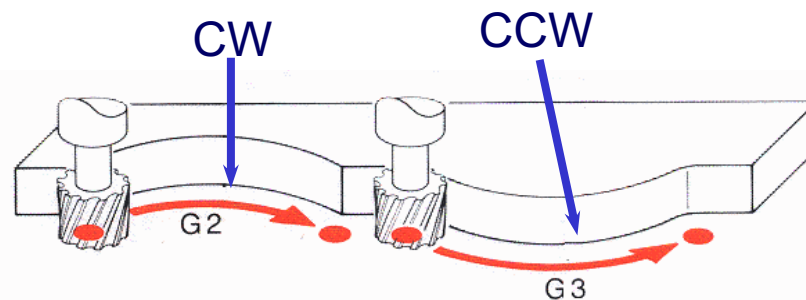
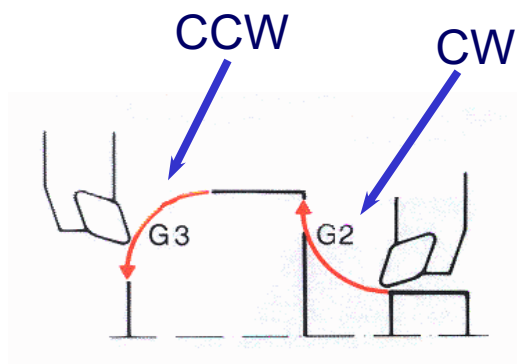
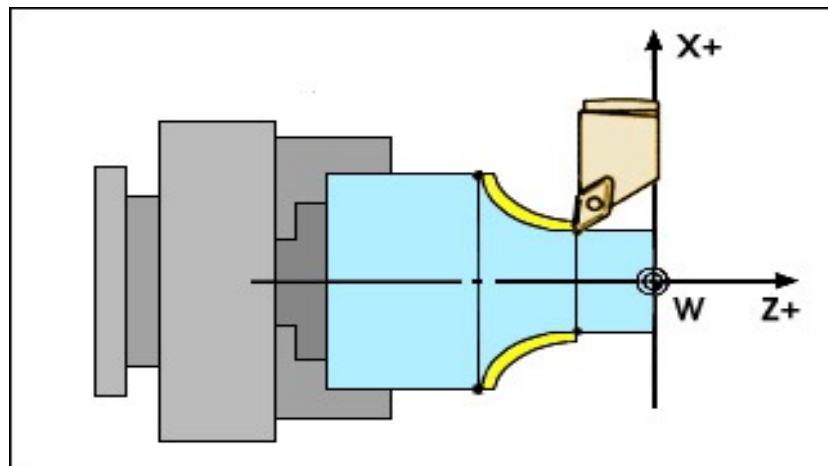
N180 G00 X20 Z2

N190 G01 Z-15 F0.2

N200 G01 X40 Z-25



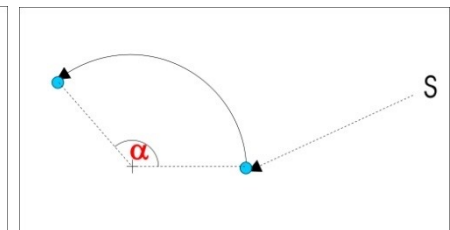
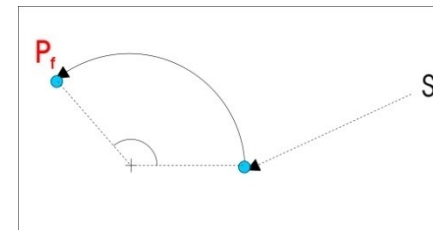
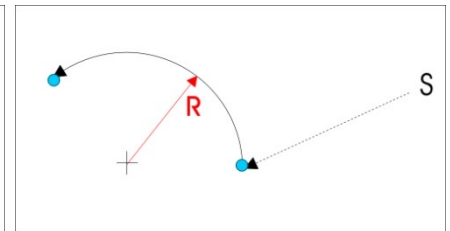
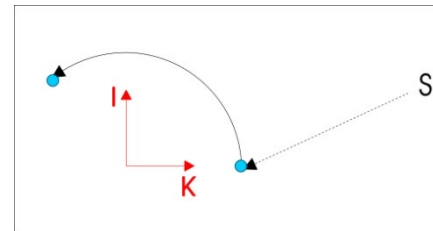
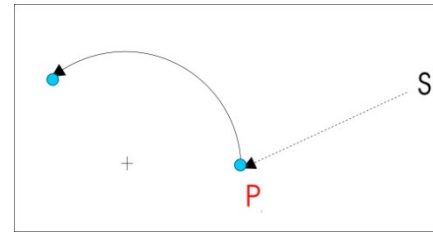
- G02 CW
- G03 CCW





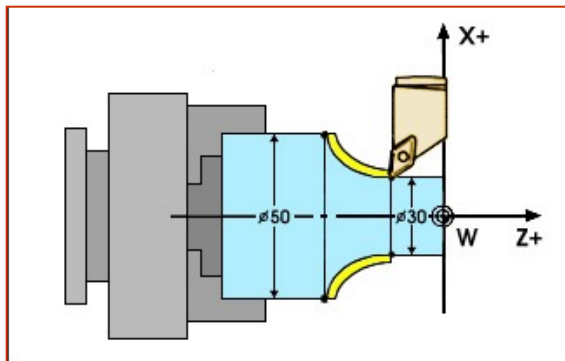
CIRCULAR INTERPOLATION: SYNTAX

1. Initial point coordinates
2. Choice between:
 - Center coordinates
plane x,y (G17): I,J
plane x,z (G18): I,K
plane y,z (G19): J,K
 - Radius R
3. Choice between:
 - Final point
 - Angle range

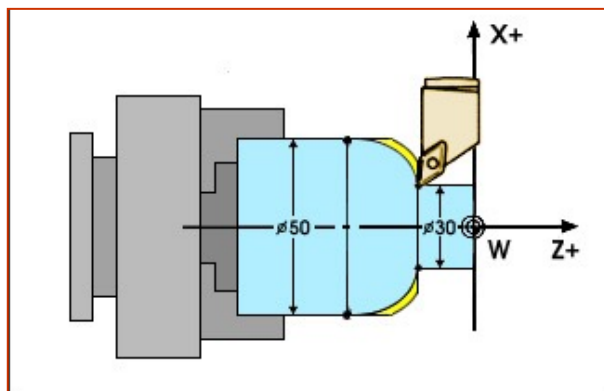
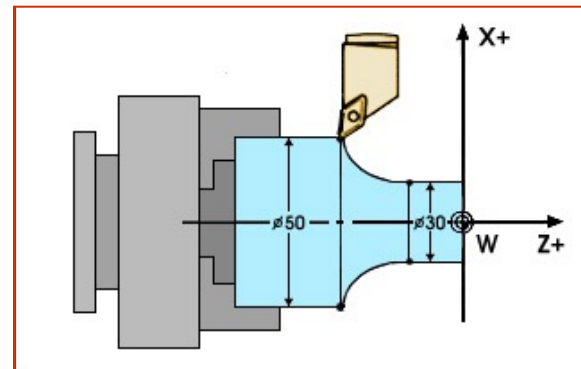




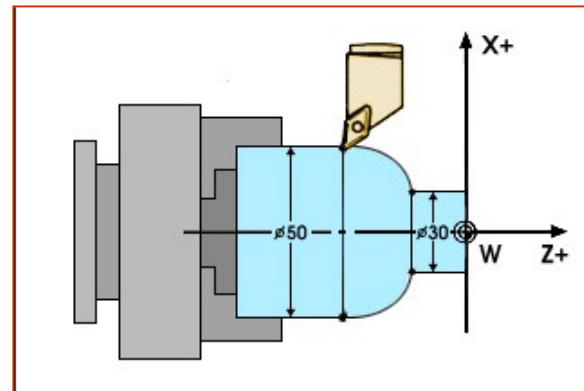
CIRCULAR INTERPOLATION: EXAMPLES



N100 **G02** X50 Z-35 I50 K-25

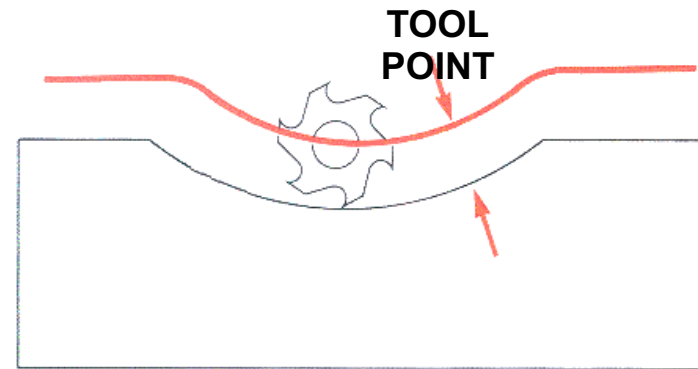
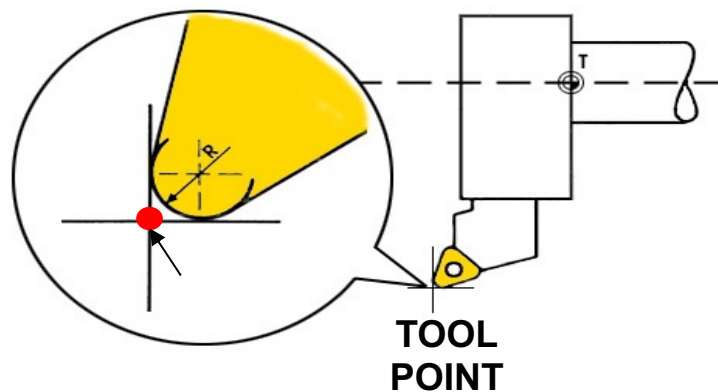


N150 **G03** X50 Z-35 I30 K-35





The tool path is programmed referring to a specific tool point.

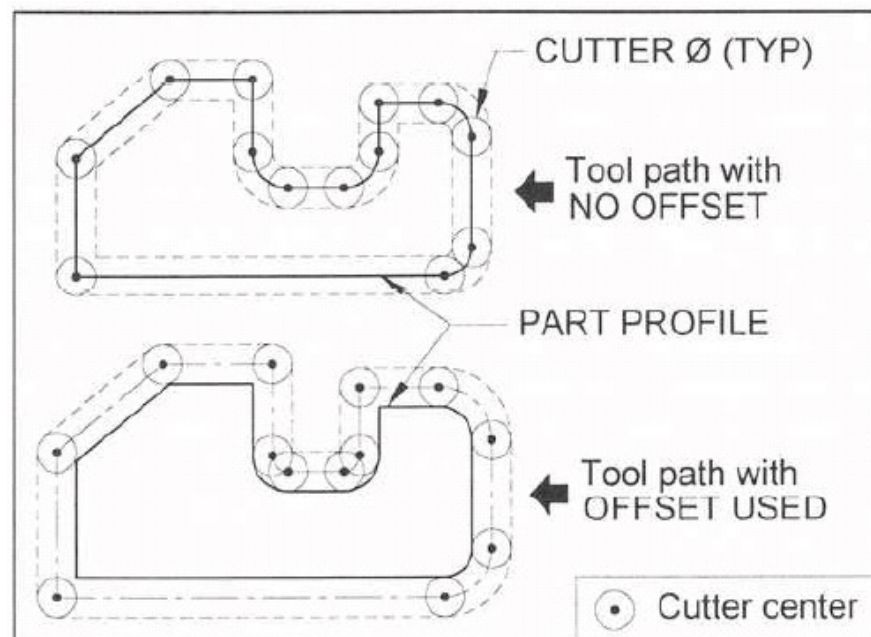




In several machining operations we have to make distinction between:

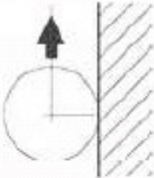
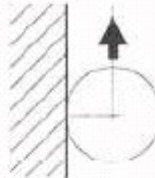
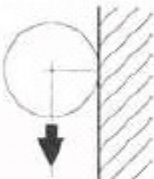
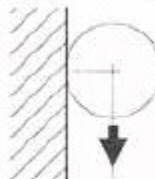
- tool path
- distances in the technical drawing

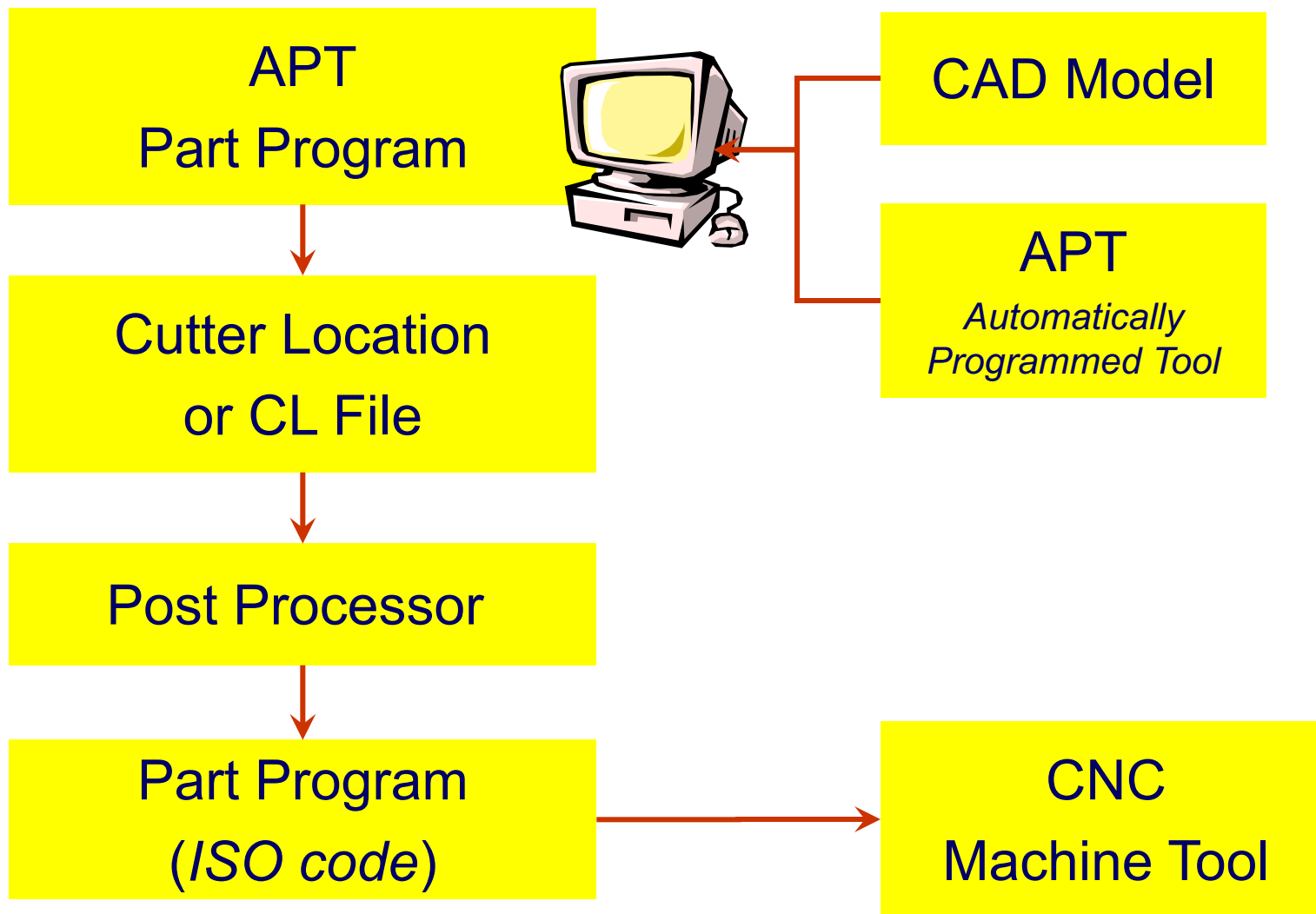
The tool compensation (or cutter radius offset) helps to program using the information in the technical drawing.





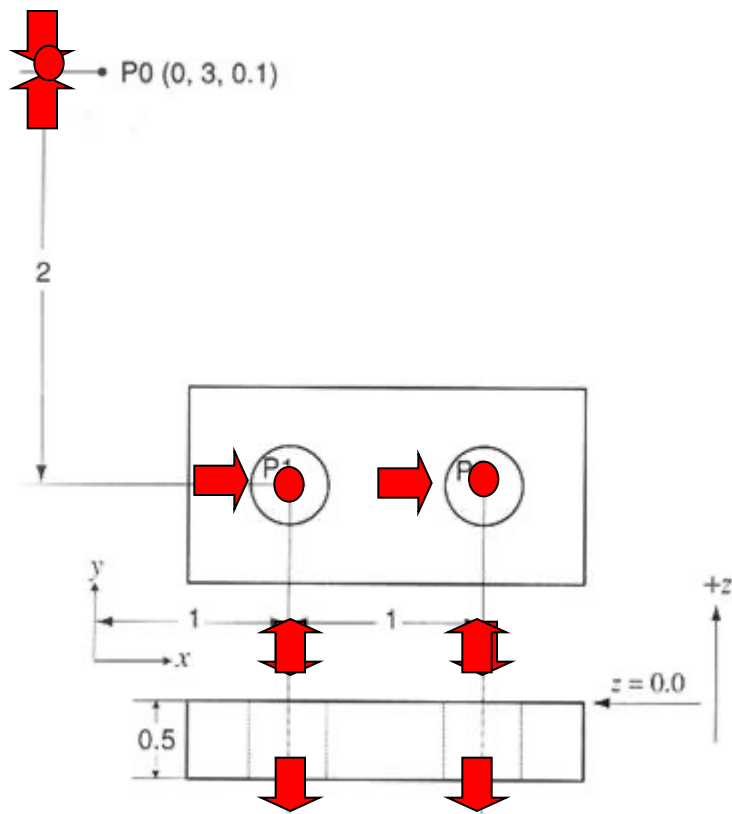
The NC calculates the trajectory of the tool point on the basis of the milling tool radius R .

	G41 LEFT	G42 RIGHT	
	G42 RIGHT	G41 LEFT	
	G40 - NONE -	G40 - NONE -	





DRILLING



```
P0=POINT/0.0,3.0,0.1  
P1=POINT/1.0,1.0,0.1  
P2=POINT/2.0,1.0,0.1  
FROM/P0  
GOTO/P1  
GODLTA/0,0,-0.7  
GODLTA/0,0,0.7  
GOTO/P2  
GODLTA/0,0,-0.7  
GODLTA/0,0,0.7  
GOTO/P0
```




Other specs:

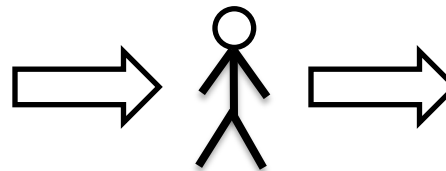
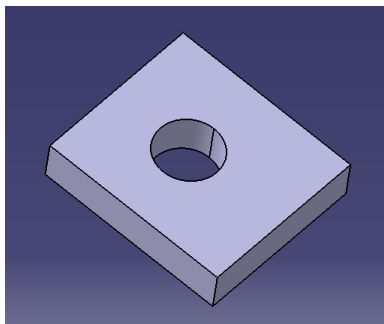
- MACHIN/ name of the machine tool
- COOLNT/ coolant on/off
- FEDRAT/ feed rate
- SPINDL/ rpm
- TOOLNO/ tool number
- CUTTER/ tool compensation
- INTOL/ o OUTTOL/ tolerance
- END (program end)



Usually, the creation/modification of a *part program* can be done in two ways:

- manually on the machine;
- using a Computer Aided Manufacturing software.

The programming on the **machine** is used when the part to be produced is simple.



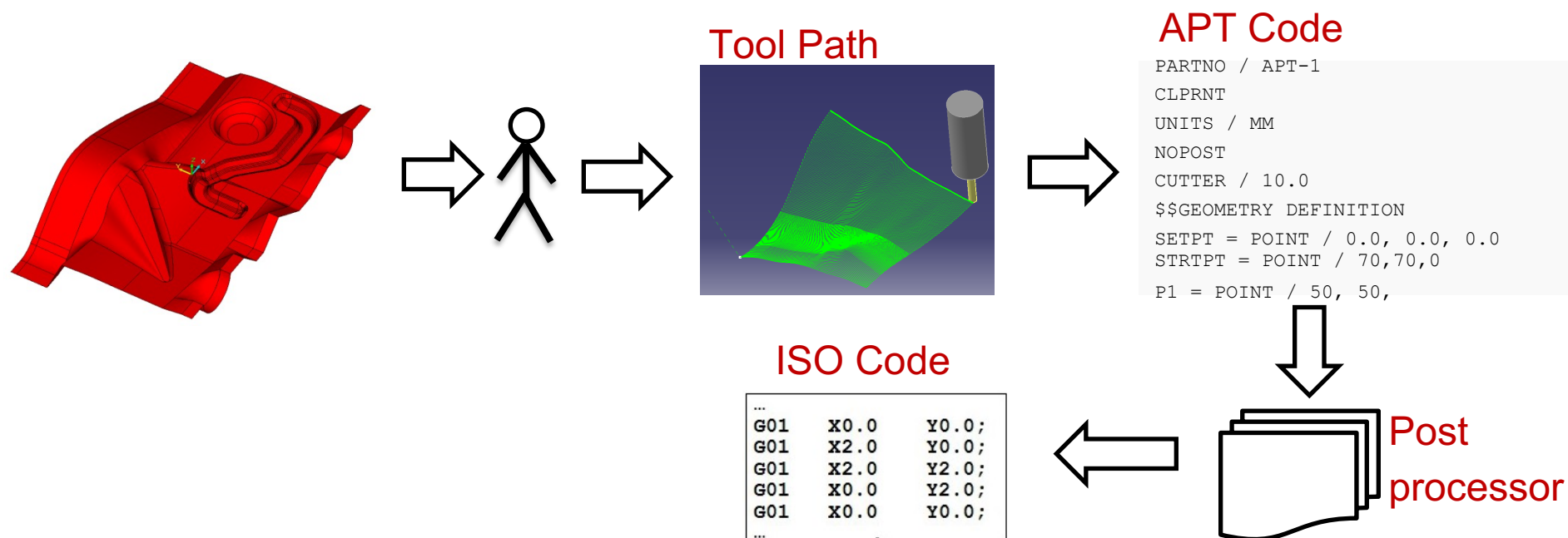
ISO Code

```
...  
G01 X0.0 Y0.0;  
G01 X2.0 Y0.0;  
G01 X2.0 Y2.0;  
G01 X0.0 Y2.0;  
G01 X0.0 Y0.0;  
...
```



CNC PROGRAMMING: CAD/CAM

The usage of a **CAM** software happens when the part is hard to manufacture, it has complicate features or requires the interaction of multiple programmers.

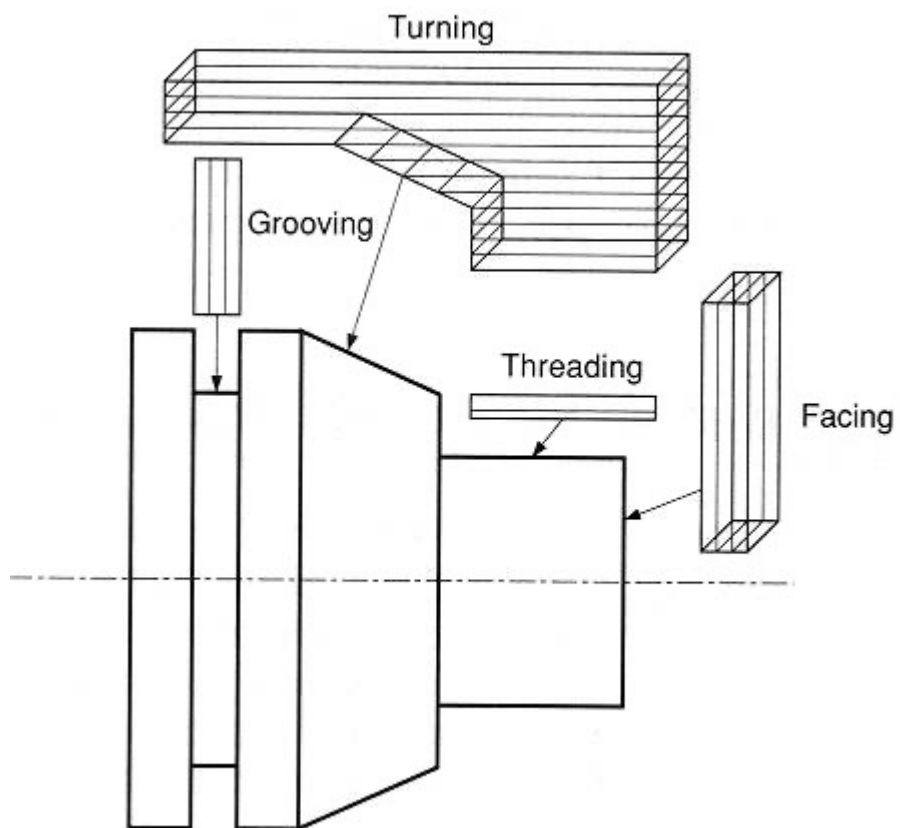




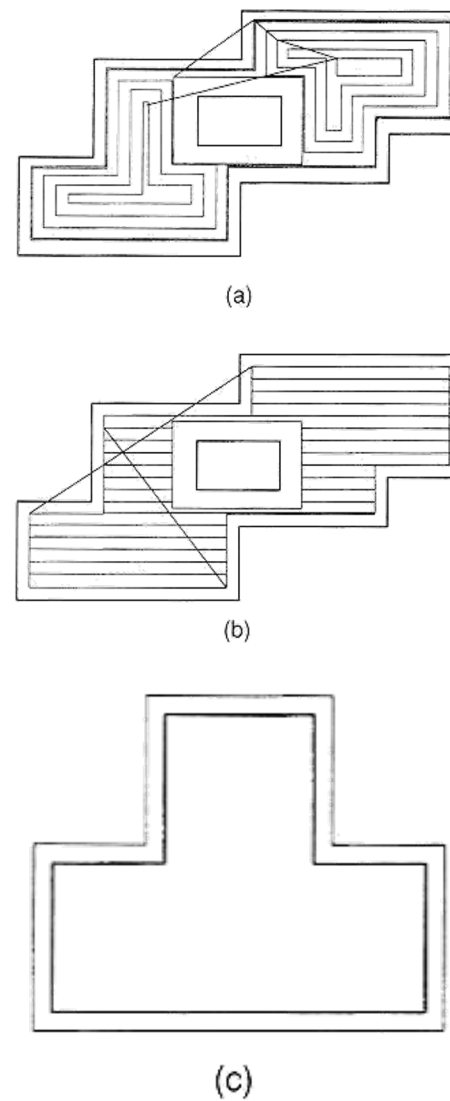
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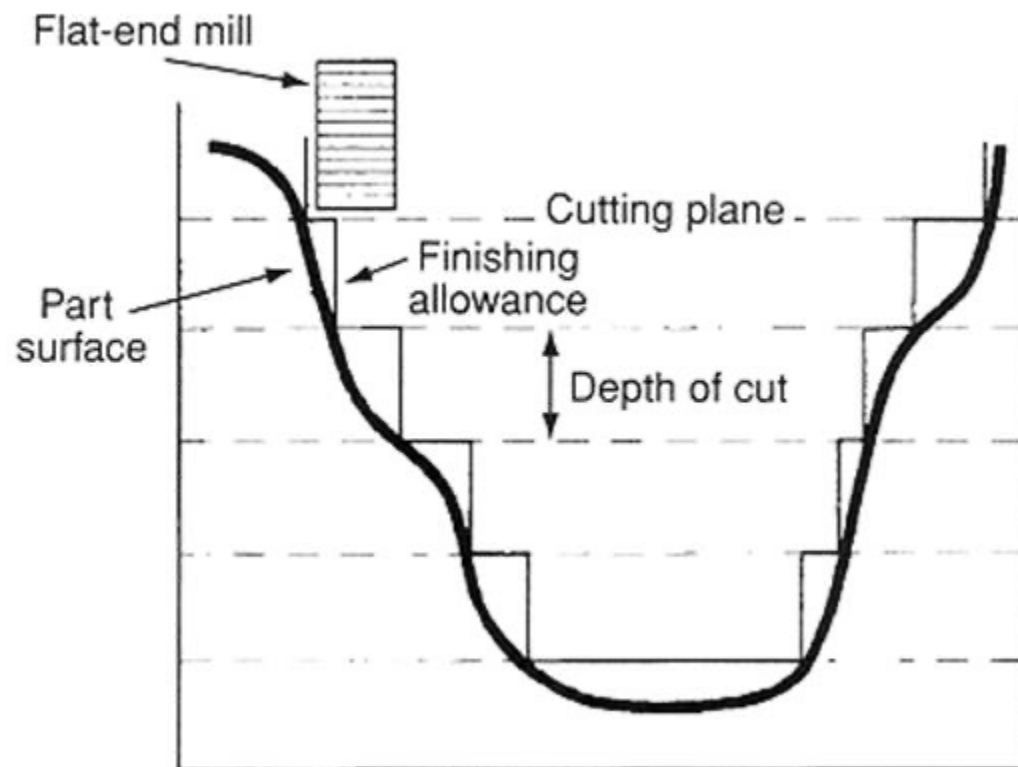
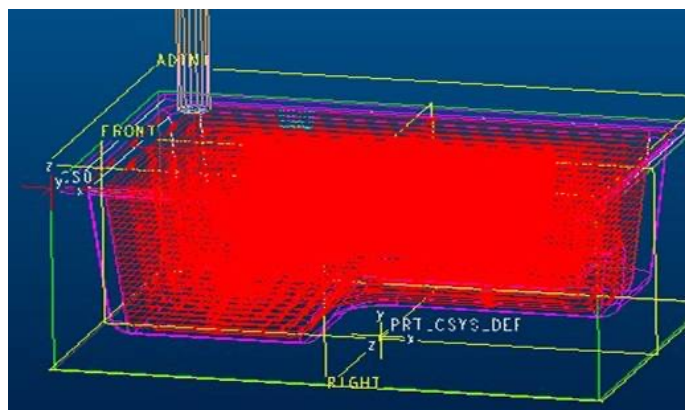
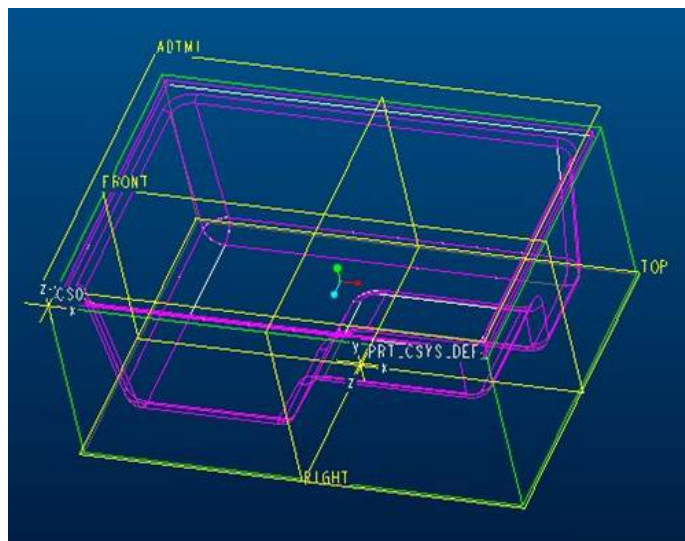


Turning



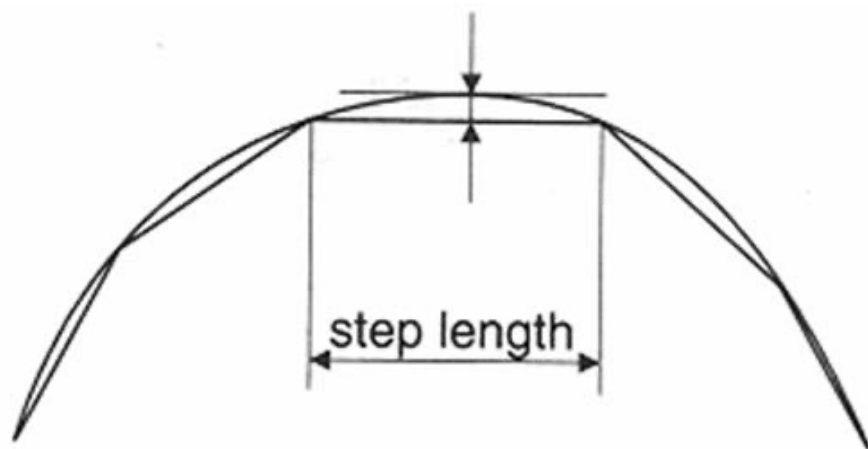
Milling



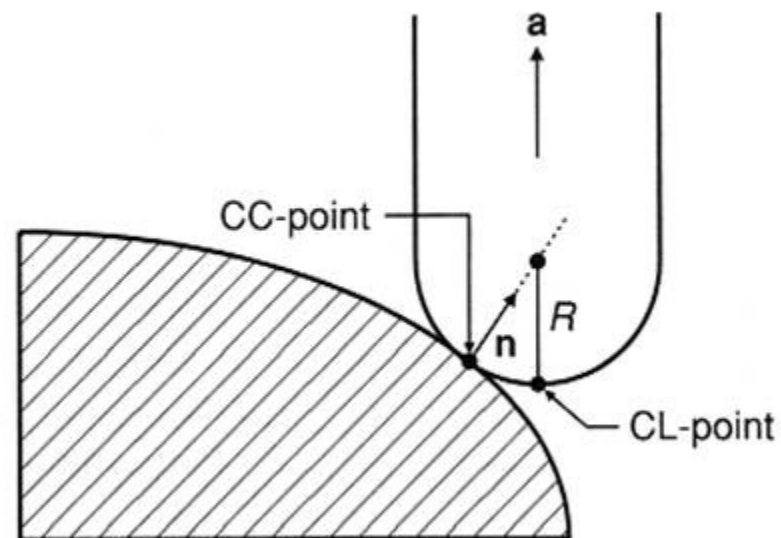




Tool path approximation

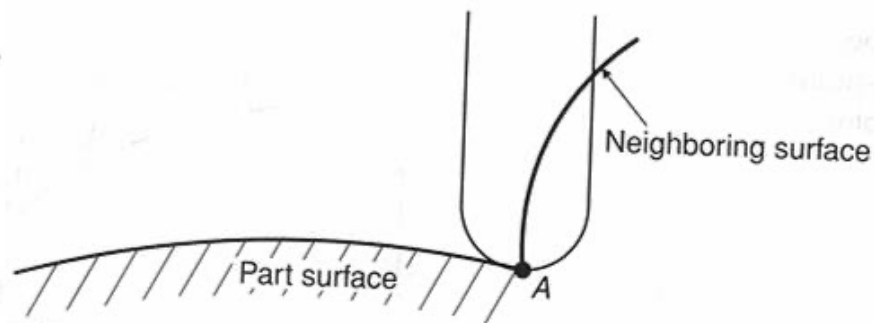
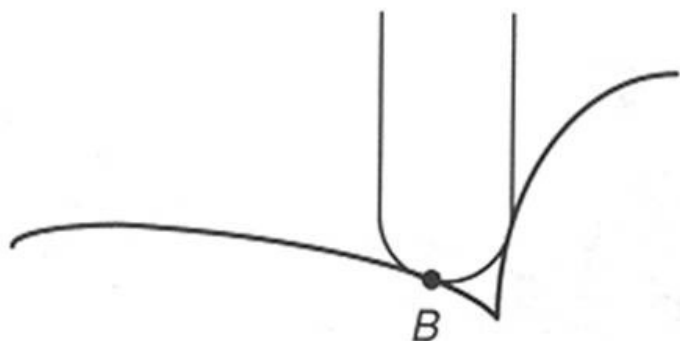
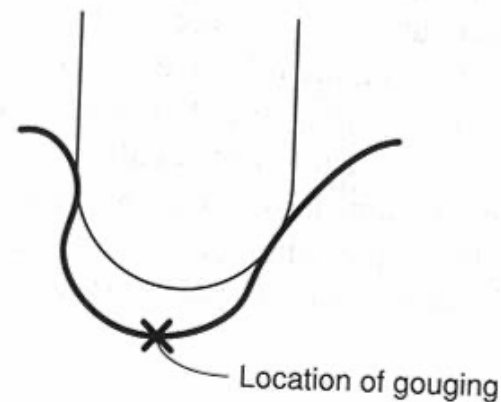
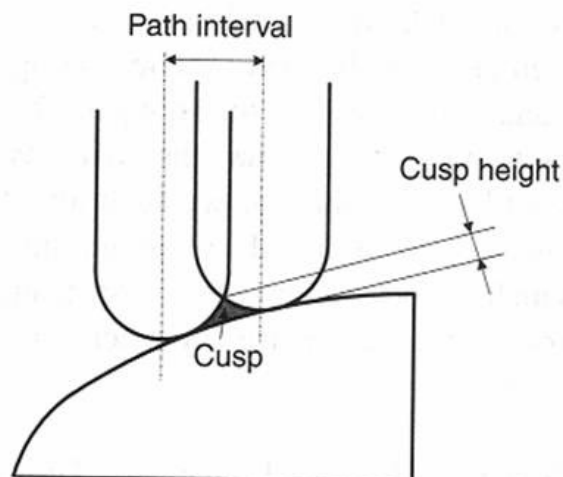


Tool compensation



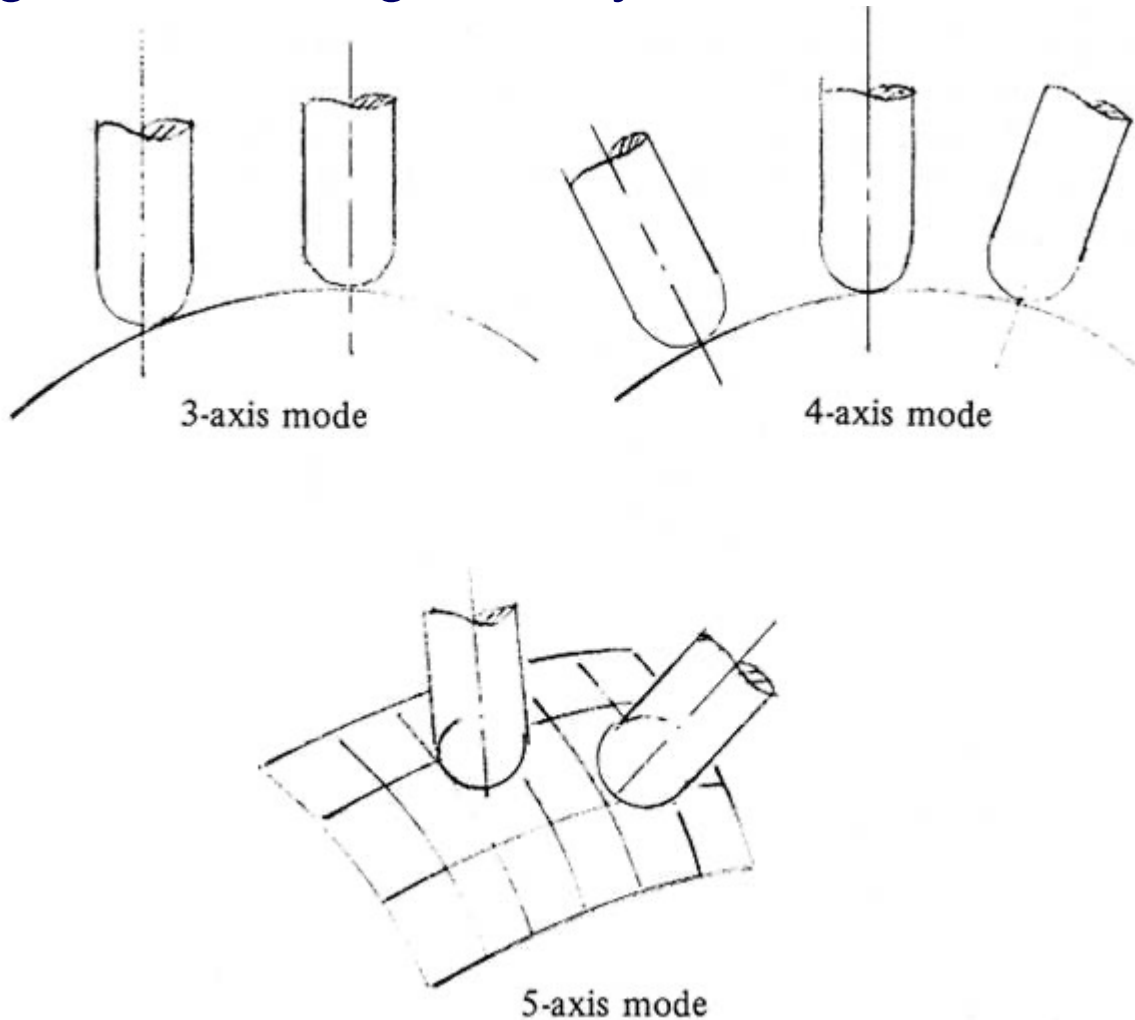


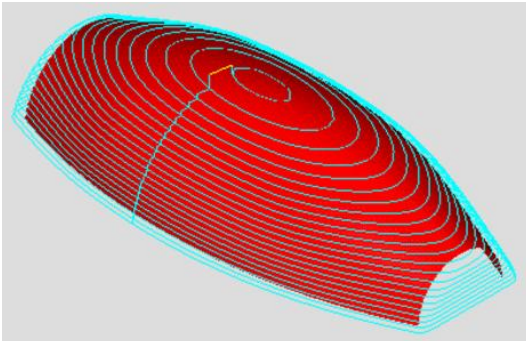
Problems in tool path generation



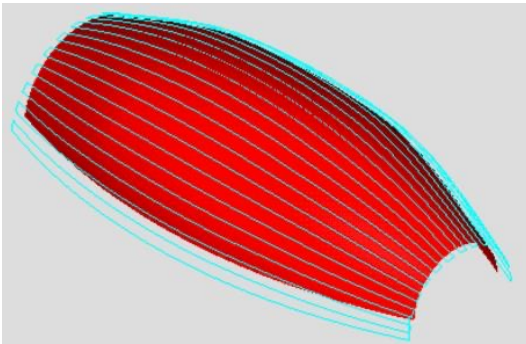


Milling of freeform geometry

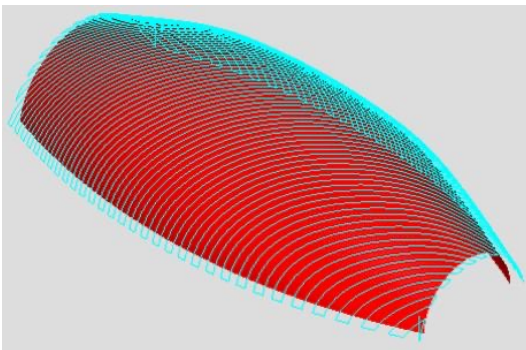




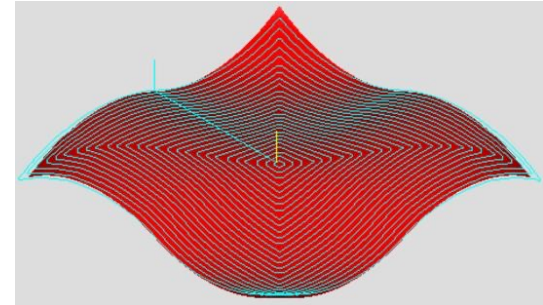
Z level
curves



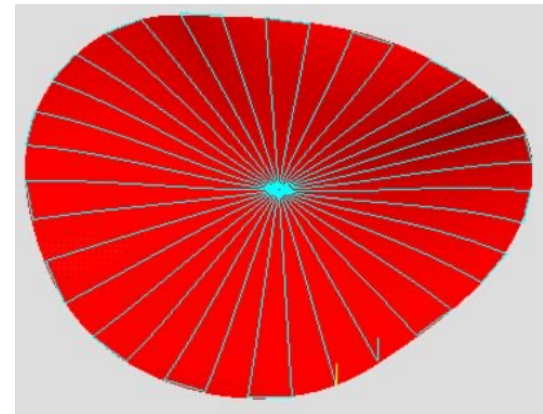
constant
parameter
curves



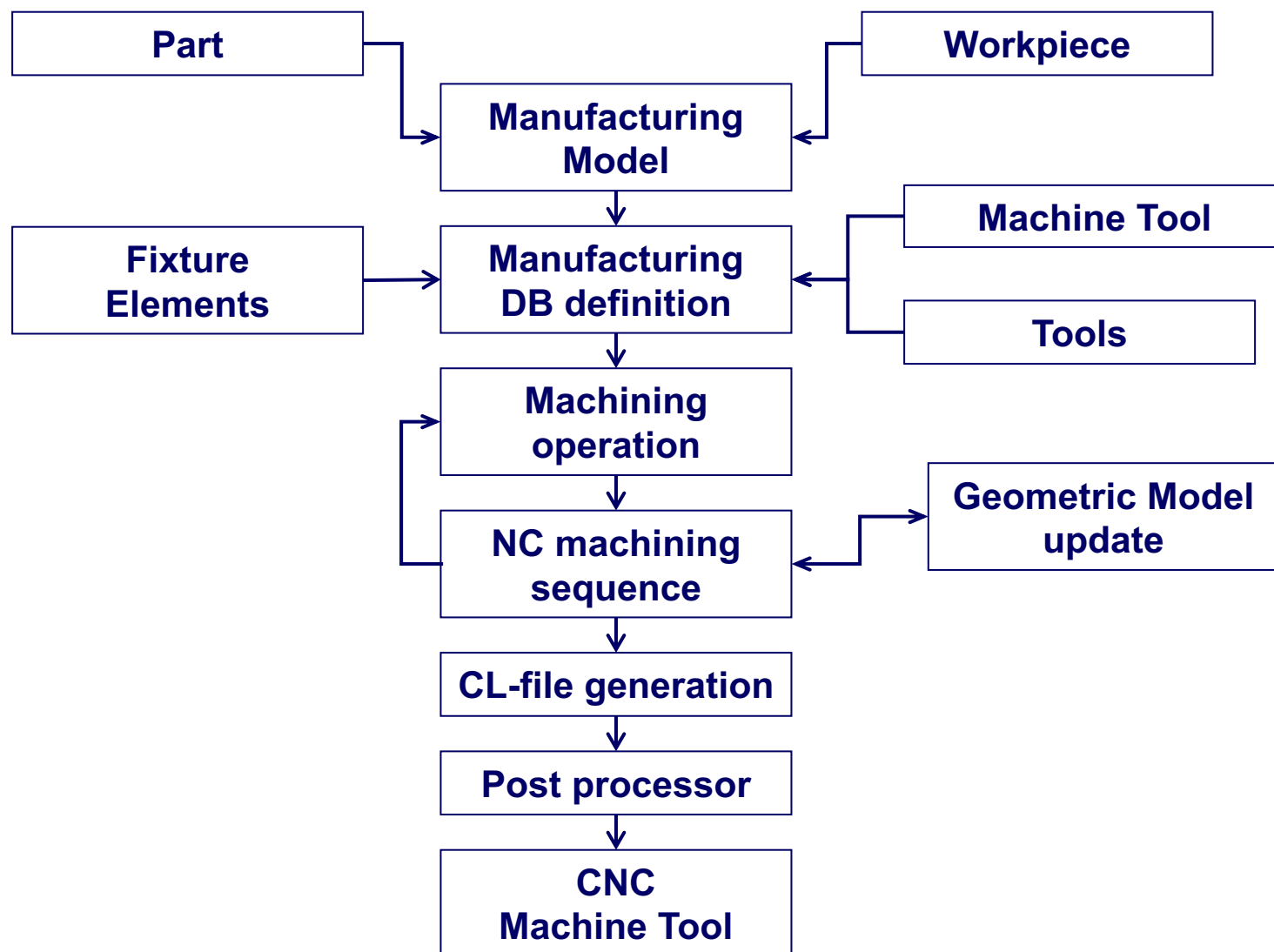
parallel
plane
curves



spiral path



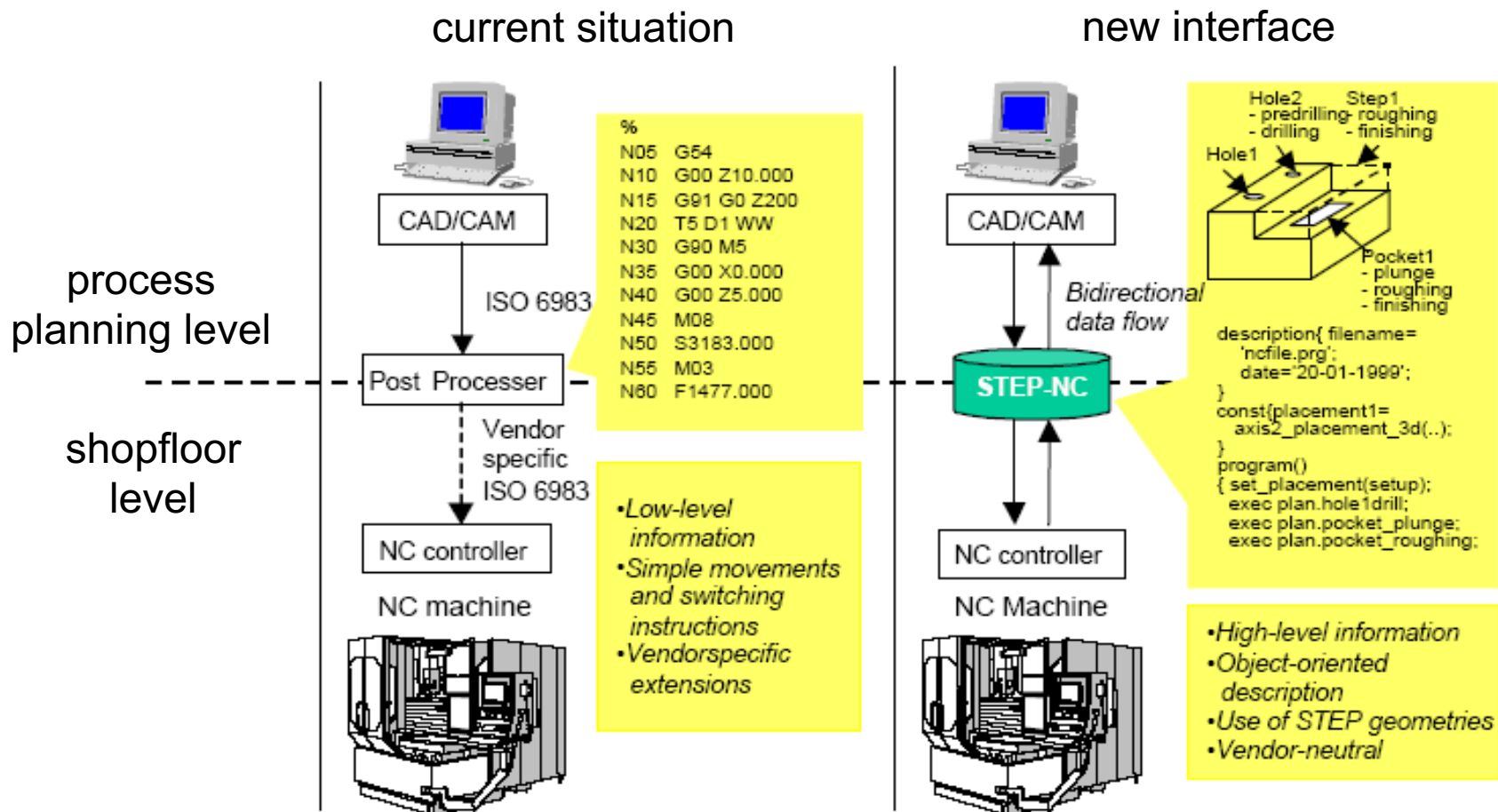
radial path





STEP-NC (ISO 14649)

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STEP-NC changes the way that manufacturing is done by defining data as "***working steps***": ***a library of specific operations that might be performed on a CNC machine tool.***

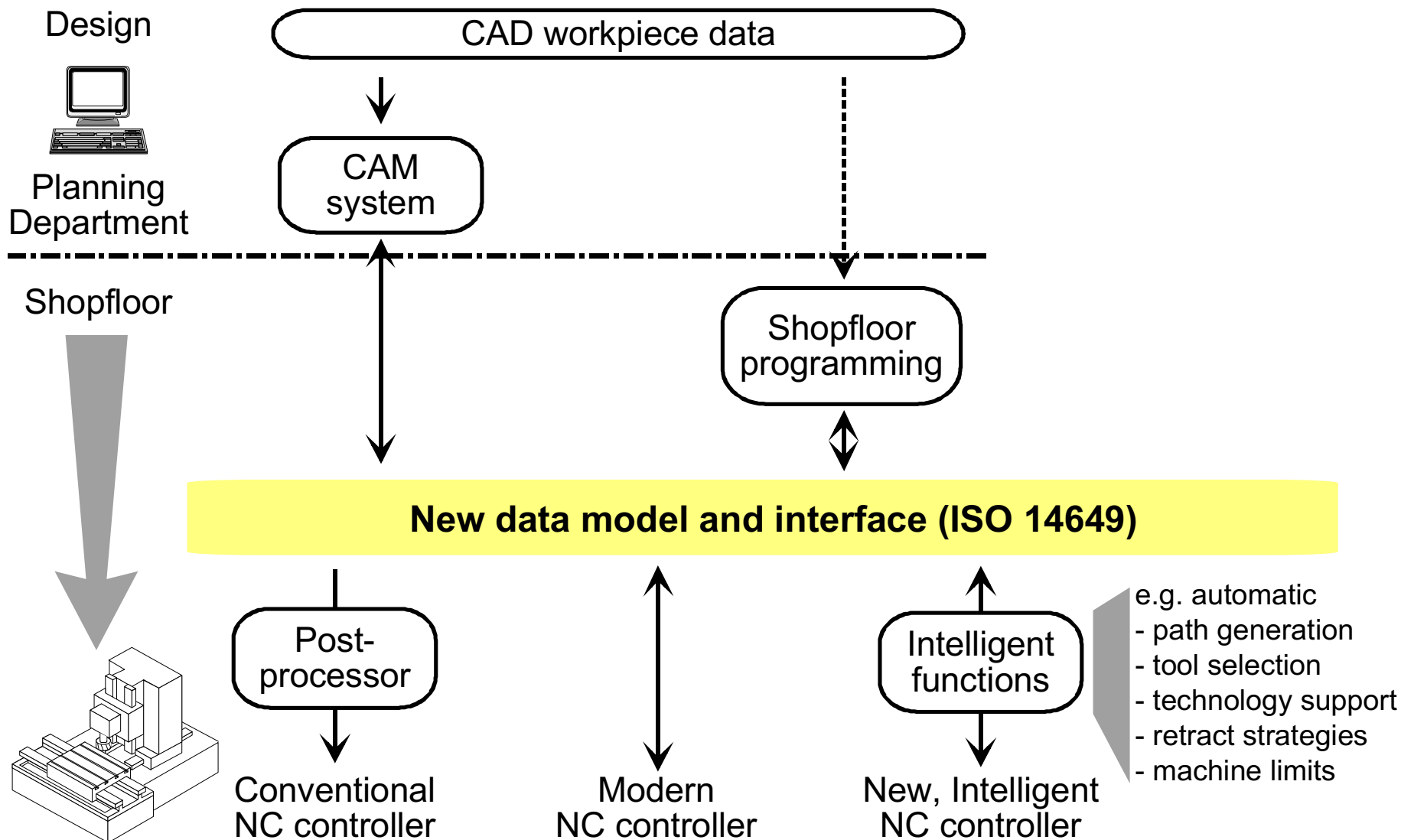
In other words, it breaks down every machining operation into the steps required to perform the operation.

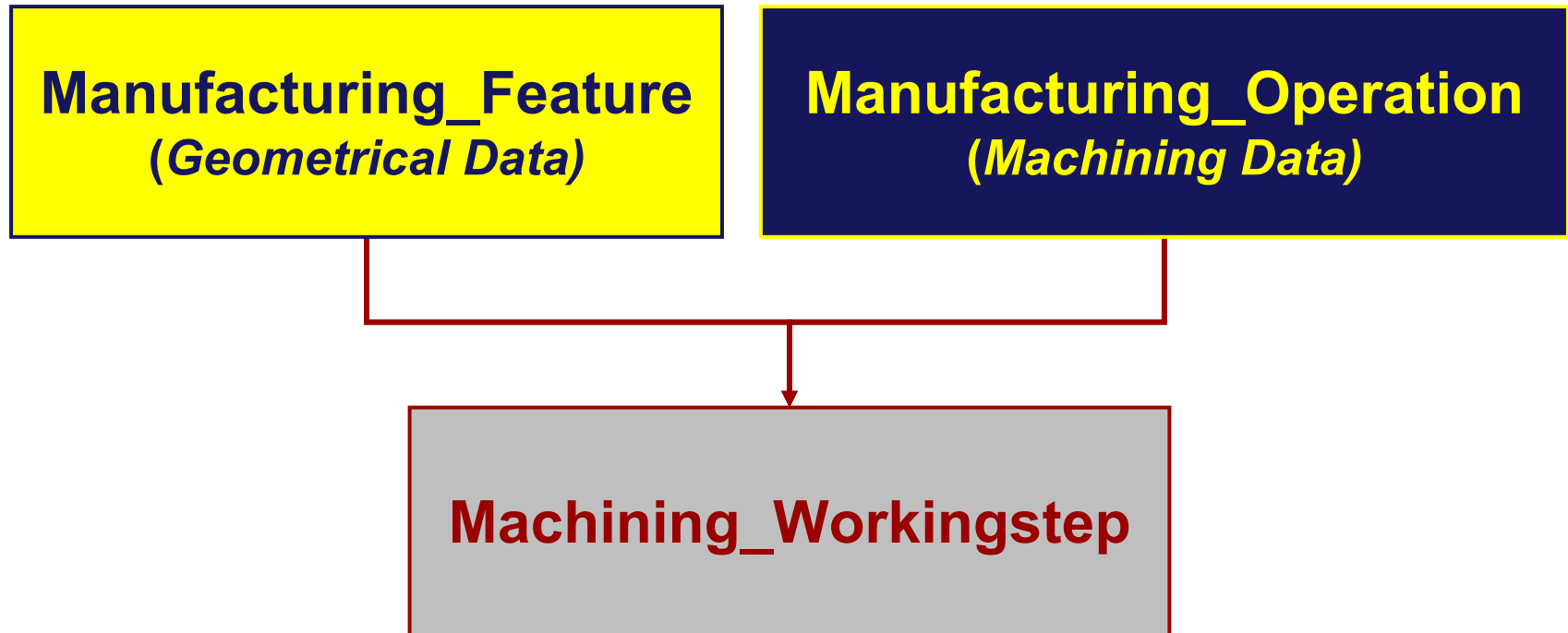
In the past, CNC machine tools had to be programmed using G and M codes (from ISO 6983), which were instructions that only told the machine what moves to make, without any semantic content that referred to the part being processed.



STEP-NC (ISO 14649)

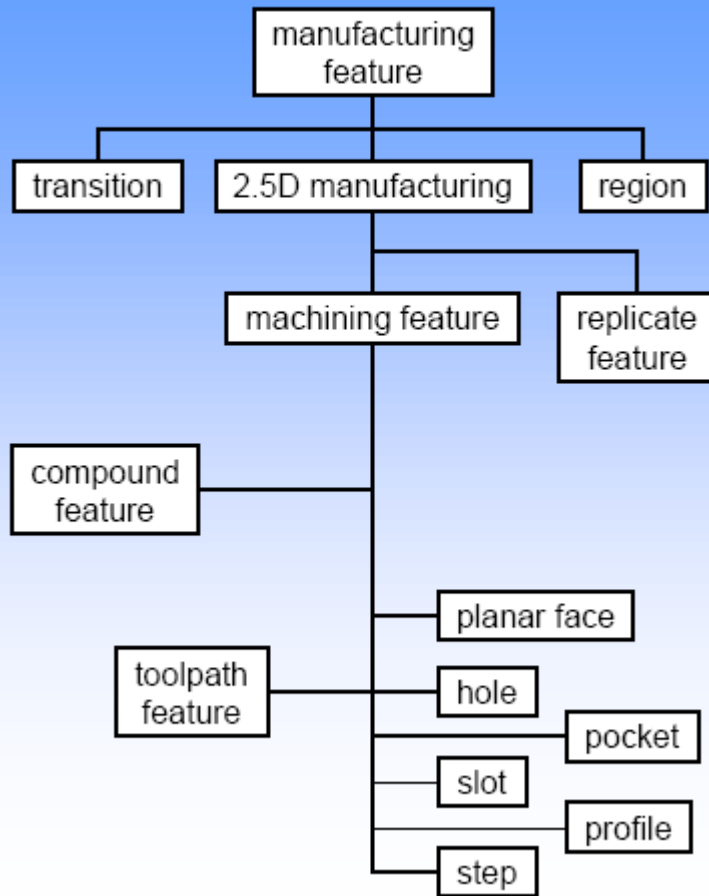
46



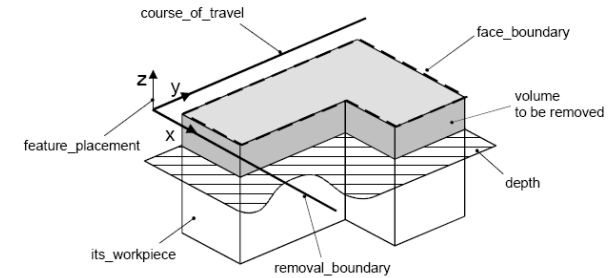




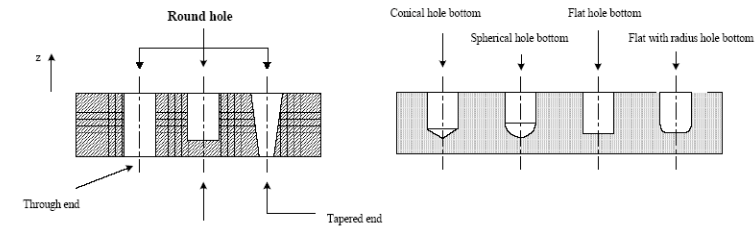
STEP-NC Features



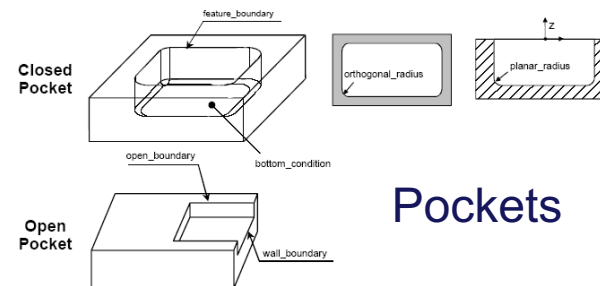
Planar Face



Round Holes

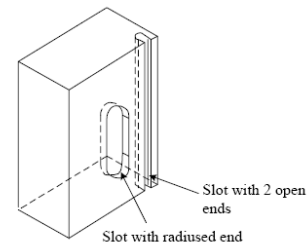


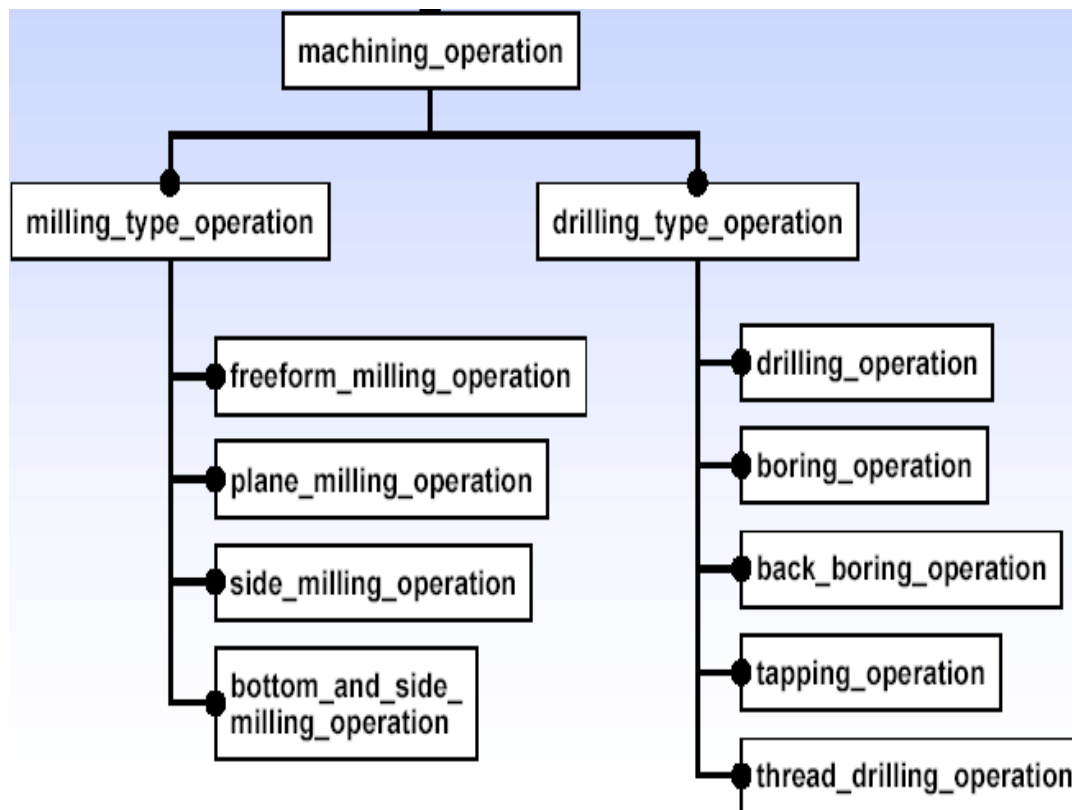
Hole Taper and Bottom Conditions



Pockets

Slots

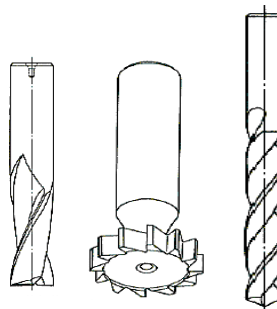




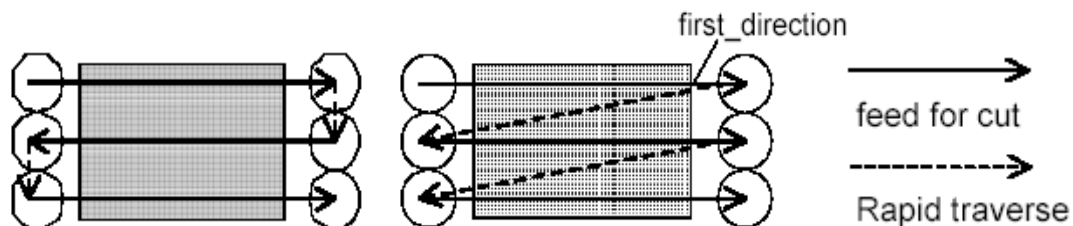


Machining Operation

Tool Requirement



Machining Strategy



Technology Parameters

Spindle Speed [RPM], Feed [mm/min]

Machine Functions

M Functions: Coolant ON/OFF.....

